

- Q.5 Work done by centripetal force is
 a) >0 b) 0
 c) <0 d) None of the above
- Q.6 When a torque acting on a system is zero, then which of the following should not change?
 a) Linear Velocity b) Angular Momentum
 c) Angular Displacement
 d) Force acting on the body
- Q.7 The heat flow through solids only by
 a) Conduction b) Convection
 c) Radiation d) Does not flow
- Q.8 The speed of the sound in a medium depends upon
 a) its elasticity as well as its inertia
 b) its elasticity but not its inertia
 c) its inertia but not its elasticity
 d) neither its elasticity nor its inertia
- Q.9 The factor that helps to recognize a person by his voice is.
 a) intensity b) pitch
 c) loudness d) quality
- Q.10 The core of a solenoid uses
 a) copper b) soft iron
 c) aluminum d) none of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 State the principle of homogeneity of dimensions.
 Q.12 Define centripetal acceleration.
 Q.13 Define work
 Q.14 Define moment of inertia.
 Q.15 Define conduction of heat transfer in solids.
 Q.16 Define emissivity.
 Q.17 Define reverberation.
 Q.18 Define refractive index.
 Q.19 Define dielectric substances.
 Q.20 Define magnetic field.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Check the accuracy of the equation $v=u+at$ by using dimensional analysis.
 Q.22 Explain the resolution of force.
 Q.23 Define friction. Give its applications in daily life.
 Q.24 Derive expression for potential energy when a body of mass (m) is lifted to a height (h).
 Q.25 Write note on free and forced vibrations.
 Q.26 Explain working of pyrometers of temperature measurement.
 Q.27 Calculate co-efficient of thermal conductivity by using Searle's method.